Euclid's Elements — Book I
Proposition 18

CGJteamLab

In any triangle the angle opposite the greater side is greater.

1 Introduction

Euclid's Proposition I.18 states that in a triangle the greater side is opposite the greater angle. For a nondegenerate triangle ABC, if

$$AB < AC,$$

then

$$\angle ACB < \angle ABC.$$

The classical proof is short but structurally rich. A copy of the shorter side AB is laid off on the longer side AC. This produces a point D with

$$A - D - C \qquad \text{and} \qquad AD \cong AB.$$

The auxiliary triangle ABD is therefore isosceles. Proposition I.5 identifies its base angles, while Proposition I.16 compares one of these angles with an angle of the original triangle. Finally, the fact that an interior subangle is smaller than the whole angle completes the comparison.

The formal reconstruction follows this synthetic argument closely. It does not introduce numerical angle measures. Instead, it uses the strict comparison relation `HilbertAngleLess` already developed for Proposition I.16.

The reconstruction also isolates two general facts about strict angle comparison:

`hilbert_interior_angle_less`

and

`hilbert_angleLess_trans`.

These belong naturally to the Hilbert interface rather than to Proposition I.18 itself.

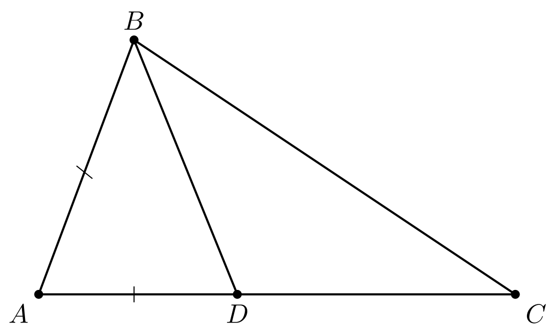
2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$AB < AC.$$

Choose D on AC so that

$$A - D - C \qquad \text{and} \qquad AD \cong AB.$$



The construction creates two auxiliary triangles.

First, triangle ABD is isosceles because

$$AB \cong AD.$$

Hence Proposition I.5 gives

$$\angle ABD \cong \angle ADB.$$

Second, consider triangle BCD. Since $A - D - C$, the ray DA is the continuation of DC through D. Thus $\angle BDA$ is an exterior angle of triangle BCD.

Proposition I.16 therefore gives

$$\angle BCD < \angle BDA.$$

Because A, D, C are collinear with D between A and C, the rays CD and CA coincide. Hence

$$\angle BCD = \angle BCA.$$

Combining these observations gives

$$\angle BCA < \angle BDA \cong \angle ABD.$$

Finally, the ray BD lies inside $\angle ABC$, since it meets the segment AC at the interior point D. Therefore

$$\angle ABD < \angle ABC.$$

By transitivity,

$$\angle BCA < \angle ABC,$$

which is the desired conclusion.

3 Classical Proof

Assume that in triangle ABC,

$$AC > AB.$$

Choose a point D on AC such that

$$AD = AB.$$

Since $AC > AB$, the point D lies strictly between A and C.

The triangle ABD is isosceles, so by Proposition I.5,

$$\angle ABD = \angle ADB.$$

Now $\angle ADB$ is an exterior angle of triangle BCD. Proposition I.16 gives

$$\angle ADB > \angle BCD.$$

Since D lies on AC,

$$\angle BCD = \angle BCA.$$

Consequently,

$$\angle ABD > \angle BCA.$$

But BD lies inside $\angle ABC$, so

$$\angle ABC > \angle ABD.$$

Therefore

$$\angle ABC > \angle BCA.$$

The classical dependency pattern is thus

$$\begin{gathered} AB < AC \\ \downarrow \\ A - D - C, \quad AB \cong AD \\ \downarrow \\ \text{I.5} \\ \downarrow \\ \angle ABD \cong \angle ADB \\ \downarrow \\ \text{I.16 on } BCD \\ \downarrow \\ \angle BCA < \angle ADB \\ \downarrow \\ \angle BCA < \angle ABD \\ \downarrow \\ \angle ABD < \angle ABC \\ \downarrow \\ \angle BCA < \angle ABC. \end{gathered}$$

4 Strict Angle Comparison

The formal statement uses the synthetic relation

$$\texttt{HilbertAngleLess}.$$

No numerical measure of angles is introduced. Informally,

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

This relation was already required for Proposition I.16. Proposition I.18 reveals that two further structural properties are needed if strict angle comparison is to behave as a genuine order relation.

The first is that an interior subangle is smaller than the whole angle.

```
        hilbert_interior_angle_less
```

Geometrically, if a ray from the vertex of $\angle AOB$ meets the opposite side AB internally, then the angle cut off by that ray is strictly smaller than $\angle AOB$.

The second property is transitivity.

```
        hilbert_angleLess_trans
```

It expresses the expected implication

$$\alpha < \beta \quad \text{and} \quad \beta < \gamma \quad \Longrightarrow \quad \alpha < \gamma.$$

Both results are independent of Proposition I.18 and were therefore added to the general Hilbert interface.

5 Proof Architecture of the Reconstruction

The formal proof separates naturally into five geometric stages.

5.1 Recovering the Interior Point

The hypothesis

$$AB < AC$$

is represented by `HilbertSegmentLess`.

Unpacking it produces a witness D satisfying

$$A - D - C$$

and

$$AB \cong AD.$$

Thus the first step of Euclid's construction is already contained in the definition of strict segment comparison.

5.2 The Isosceles Triangle

From

$$AB \cong AD$$

the existing theorem

`hilbert_isosceles_base_angles`

gives

$$\angle ABD \cong \angle ADB.$$

This is precisely Proposition I.5 in the form required by the reconstruction.

5.3 The Exterior-Angle Comparison

Since

$$A - D - C,$$

we also have

$$C - D - A.$$

Apply Proposition I.16 to triangle BCD with DA as the extension of DC. This yields

$$\angle BCD < \angle BDA.$$

The ray from C through D is the same ray as the ray from C through A. Hence the first angle can be normalized:

$$\angle BCD = \angle BCA.$$

Thus

$$\angle BCA < \angle BDA.$$

Using the isosceles-angle congruence and the symmetry of the unordered angle representation gives

$$\angle BCA < \angle ABD.$$

5.4 The Interior Subangle

The point D lies on the segment AC. Consequently the ray BD meets the interior of that segment.

Therefore BD lies inside $\angle ABC$, and `hilbert_interior_angle_less` gives

$$\angle ABD < \angle ABC.$$

5.5 Transitivity

We now have

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC.$$

Applying `hilbert_angleLess_trans` gives

$$\angle BCA < \angle ABC.$$

The final public statement writes the first angle as $\angle ACB$. Since the formal angle representation is insensitive to reversal of its two rays, this is a final orientation normalization rather than an additional geometric argument.

6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_18
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C : Geo.Point)
(hABC : not PrimCollinear Geo A B C)
(hABltAC : HilbertSegmentLess Geo A B A C) :
HilbertAngleLess Geo A C B A B C := by
```

The hypothesis `hABltAC` is immediately unpacked:

```
rcases hABltAC with
<D, hADC, hAB_AD>
```

Thus `hADC` records

$$A - D - C,$$

while `hAB_AD` records

$$AB \cong AD.$$

The isosceles part of the argument is then:

```
have hIso :
Geo.AngleCongruent A B D A D B :=
hilbert_isosceles_base_angles
Geo
A B D
hABD
hAB_AD
```

The exterior-angle step uses Proposition I.16:

```
have hBCD_BDA :
HilbertAngleLess Geo B C D B D A :=
euclid_proposition_16_second
Geo B C D A hBCD hCDA
```

Here

$$C - D - A$$

is obtained from the original betweenness relation

$$A - D - C.$$

The ray identity $CD = CA$ is expressed by `HilbertSameRay`:

```
have hRayCDA :
HilbertSameRay Geo C D A :=
hilbert_sameRay_of_between
Geo C D A hCDA
```

This permits normalization of

$$\angle BCD$$

to

$$\angle BCA.$$

After transporting the right-hand angle across the isosceles congruence, the proof obtains

```
have hBCA_ABD :
HilbertAngleLess Geo B C A A B D := ...
```

which represents

$$\angle BCA < \angle ABD.$$

For the second comparison, the ray BD meets the segment AC at D:

```
have hInsideBD :
HilbertRayMeetsSegment Geo B D A C :=
<D, hADC, hRayBDD>
```

Hence

```
have hABD_ABC :
HilbertAngleLess Geo A B D A B C :=
hilbert_interior_angle_less
Geo
B D A C
hABC
hInsideBD
```

which is exactly

$$\angle ABD < \angle ABC.$$

Transitivity now gives

```
have hFinal :
HilbertAngleLess Geo B C A A B C :=
hilbert_angleLess_trans
Geo
B C A
A B D
A B C
hBCA_ABD
hABD_ABC
```

The final lines merely reverse the first angle from

$$\angle BCA$$

to

$$\angle ACB,$$

matching the public orientation of the proposition.

7 Relationship with Earlier Propositions

7.1 Proposition I.5

Proposition I.5 supplies the equality of the base angles in the auxiliary isosceles triangle ABD:

$$AB \cong AD \quad \Longrightarrow \quad \angle ABD \cong \angle ADB.$$

This is the bridge between the comparison of sides and the comparison of angles.

7.2 Proposition I.16

Proposition I.16 supplies the strict inequality. Applied to triangle BCD, it compares the remote interior angle at C with the exterior angle at D:

$$\angle BCD < \angle BDA.$$

Thus I.18 depends essentially on the exterior-angle theorem proved two propositions earlier.

7.3 Proposition I.17

Proposition I.17 is not required by the proof of I.18. The formal dependency follows Euclid's direct argument through I.5 and I.16.

This is useful structurally: I.18 does not rely on an angle-sum theory developed for I.17. Both propositions instead reuse the synthetic strict-angle relation introduced for I.16.

8 Hilbert and Book Zero Components Used

8.1 `HilbertSegmentLess`

The hypothesis that one side is shorter than another already contains the point required by Euclid's construction. Unpacking

$$AB < AC$$

produces

$$A - D - C \qquad \text{and} \qquad AB \cong AD.$$

Thus no separate segment-layoff construction is required inside Proposition I.18.

8.2 `hilbert_isosceles_base_angles`

This theorem supplies the I.5 step for the auxiliary triangle ABD.

8.3 `HilbertSameRay`

Same-ray machinery is used to recognize that, because $A-D-C$, the rays CD and CA determine the same side of the angle at C.

This turns the I.16 result about $\angle BCD$ into the required comparison involving the original angle $\angle BCA$.

8.4 `hilbert_interior_angle_less`

This new interface theorem expresses the elementary principle that a proper interior part of an angle is smaller than the whole angle.

In I.18 it gives

$$\angle ABD < \angle ABC$$

from the fact that BD meets AC internally.

8.5 `hilbert_angleLess_trans`

This new interface theorem provides transitivity of strict angle comparison.

It combines

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC$$

into the final inequality.

9 Dependency Summary

The formal dependency structure is

$$
\begin{gathered}
\text{Hilbert incidence} + \text{order} + \text{congruence} \\
\downarrow \\
\texttt{HilbertSegmentLess} \quad \texttt{HilbertAngleLess} \\
\downarrow \\
A - D - C, \quad AB \cong AD \\
\downarrow \\
\texttt{hilbert\_isosceles\_base\_angles} \quad (\text{I.5}) \\
\downarrow \\
\angle ABD \cong \angle ADB \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\angle BCA < \angle ABD \\
\downarrow \\
\texttt{hilbert\_interior\_angle\_less} \\
\downarrow \\
\angle ABD < \angle ABC \\
\downarrow \\
\texttt{hilbert\_angleLess\_trans} \\
\downarrow \\
\texttt{euclid\_proposition\_18}.
\end{gathered}
$$

No new geometric axiom is introduced.

The two results added to the Hilbert interface are structural properties of the already existing strict-angle relation rather than assumptions specific to Proposition I.18.

10 Conclusion

Proposition I.18 is a particularly clear example of how Euclid's construction survives formal reconstruction.

The classical proof introduces a point D on the longer side, creates an isosceles triangle, applies the exterior-angle theorem, and finally observes that a part of an angle is smaller than the whole. The Lean proof follows exactly this geometry.

What changes is not the mathematical argument but the amount of structure that must be made explicit. The informal steps

$$\angle BCA < \angle ABD < \angle ABC$$

require a formal notion of strict angle comparison, a theorem asserting that an interior subangle is smaller than its containing angle, and a proof of transitivity.

Consequently, Proposition I.18 does more than add another theorem of Book I. It strengthens the reusable language of synthetic angle order. The new results `hilbert_interior_angle_less` and `hilbert_angleLess_trans` are available for subsequent propositions independently of I.18.

The completed Lean development compiles cleanly and contains no `sorry`.